

Md Abdullah Al Zaman

Department of Physics and Astronomy
University of Kansas
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PROFESSIONAL EXPERIENCE

University of Kansas (KU)

Graduate Research Assistant
Graduate Teaching Assistant

01/2025 – Present
08/2024 – 12/2024

Northern University Bangladesh (NUB)

Senior Lecturer (Study Leave)
Lecturer

08/2023 – Present
09/2019 – 07/2023

EDUCATION

University of Kansas (KU)

PhD in Physics (Astronomy Track)
Advisor: Dr. Ian J. M. Crossfield, Associate Professor, KU

08/2024 – Present

University of Chittagong (CU)

M.S. in Physics

Position: 5th (Out of 75)

Advisor: Dr. Md. Rafiqul Islam, Professor, CU

Thesis Title: Study of SMES device and SMES-PCC computer program development for the analysis of superconducting coil design

01/2018 – 05/2019

B.Sc. in Physics

Department of Physics, CU

Position: 4th (Out of 92)

01/2013 – 01/2018

RESEARCH INTERESTS

- Exoplanet Detection
- Transit Photometry
- Planetary Atmospheres
- Biosignatures
- Telescope Mission Design
- Space Radiation and Shielding

PUBLICATIONS

Citations: 185 (June 2026) • h-index: 9 • Source: ResearchGate

FIRST AUTHOR

1. **Zaman, MD. A. Al**, & Crossfield, Ian J. M. (2026) Reassessment of Planetary Parameters and Statistical Validation of 188 Kepler Habitable-Zone and Near Habitable Confirmed and Candidate Planets (**In preparation**)
2. **Zaman, MD. A. Al**, & Crossfield, Ian J. M. (2026) STERNA: A Python-Based Observation Planning Tool for Exoplanet Science (**In preparation**)
3. **Zaman, MD. A. Al**, & Kunja, L. A. (2023) Effectiveness of radiation shields constructed from Martian regolith and different polymers for human habitat on Mars using MULASSIS/GEANT4 and OLTARIS. *AIP Advances*, 13(8).
4. **Zaman, MD. A. Al**, & Monira, N. J. (2023) Shielding effectiveness of different polymers and low-density hydrides in a combined radiation shield for crewed interplanetary space missions. *Radiation Physics and Chemistry*, 205, 110706.
5. **Zaman, MD. A. Al**, Roy, S., Sarker, S., et al. (2023) Comparative Analysis of the Effectiveness of Natural Polymers and Conventional Space Radiation Shielding Polymers in Spacecraft for Prolonged Space Expeditions. *Materials Research Express*.
6. **Zaman, MD. A. Al**, Islam, M. R., Maruf, H. M. A. R., & Nizam, Q. M. R. (2022) Effectiveness of Kevlar and water-soaked hygienic wipes in a combined radiation shield for manned long termed space missions. *Radiation Physics and Chemistry*, 110483.
7. **Zaman, MD. A. Al**, & Nizam, Q.M.R. (2022) Study on Shielding Effectiveness of a Combined Radiation Shield for Manned Long Termed Interplanetary Expeditions. *Journal of Space Safety Engineering*.
8. **Zaman, MD. A. Al**, Maruf, H. M. A. R., & Islam, M. R. (2021) Study on superconducting magnetic shield for the manned long termed space voyages. *The Egyptian Journal of Remote Sensing and Space Science*, 24(2), 203-210.
9. **Zaman, MD. A. Al**, Islam, M., & Maruf, H. (2020) Study on Conceptual Designs of Superconducting Coil for Energy Storage in SMES. *East European Journal of Physics*, (1), 111-120.
10. Islam, M. R., & **Zaman, MD. A. Al** (2020) SMES-PCC computer program development for studying and designing the superconducting magnets for energy storage purpose. *Solid State Science and Technology*, 27(1 & 2), 40-50.

CONTRIBUTING AUTHOR

1. Roy, A., **Zaman, MD. A. Al** (2026) The Lazuli Space Observatory: Architecture and Capabilities (arxiv.org/abs/2601.02556, *Submitted*)
2. Brande, J., **Zaman, MD. A. Al** (2026) Carbon dioxide and evidence for water in a stripped Hot Jupiter (*Submitted*)

TALKS AND POSTERS

CONFERENCE TALKS

1. **Zaman, MD. A. Al**, Crossfield, Ian J. M., Brande, Jonathan, Douglas, Ewan S. (2025). STERNA: A Python-based Batoid-integrated Observation Planning Tool for Exoplanet Science. *5th International Conference on Physics for Sustainable Development and Technology (ICPSDT-2025)*, October 29-30, 2025.
2. **Zaman, MD. A. Al**, Gupta, S. D., Maruf, H. M. A. R., Gupta, S. D. (2025). Estimating Low-Density Hydride Effectiveness in Spacesuit Radiation Shielding: A Comparative Computational Approach. *KU Physics & Astronomy Locally Organized Assembly (PALOOZA'25)*.
3. **Zaman, MD. A. Al**, & Kunja, L. A. (2024). Radiation Shielding Strategies for Human Habitats on Mars: Exploring the Effectiveness of Shields Constructed with Martian Regolith and Hydrogenous Polymers Using Geant4 and OLTARIS. *Mid-American Regional Astrophysics Conference (MARAC)*, December 5-6, 2024.
4. **Zaman, MD. A. Al**, Islam, M. R., & Maruf, H. M. A. R. (2023). Star Formation Rate Profiles in Spiral Galaxies at $z < 0.03$. *5th International Conference on Physics for Sustainable Development and Technology (ICPSDT-2023)*, September 7-8, 2023.
5. Kunja, L. A., & **Zaman, MD. A. Al** (2023). Effectiveness of radiation shields constructed from Martian regolith and different polymers for human habitat on Mars using MULASSIS/GEANT4 and OLTARIS. *Korean Space Science Society (KSSS) Fall Conference*, October 25-27, 2023.
6. **Zaman, MD. A. Al**, Islam, M. R., Maruf, H. M. A. R., & Nizam, Q. M. R. (2022). Study on Effectiveness of Polyimide and Polyester Polymer for Radiation Shielding in a Spacecraft. *International Conference on Physics*, Bangladesh Physical Society.
7. **Zaman, MD. A. Al**, Maruf, H. M. A. R., & Islam, M. R. (2020). Study on Superconducting Magnetic Shield for the Manned Space Expeditions. *International Conference on Physics*, Atomic Energy Center, Dhaka.
8. **Zaman, MD. A. Al**, & Islam, M. R. (2019). Study of Design Optimization of Superconducting Solenoid and Toroid Coil for the Energy Storage in SMES. *National Conference on Physics*, Bangladesh Physical Society.

POSTERS

1. **Zaman, MD. A. Al**, & Crossfield, Ian J. M. (2026). Towards precise Eta Earth Estimates: Reassessment of planet parameters and Statistical Validation for 144 Kepler Confirmed and Candidate Planets. *American Astronomical Society Winter Meeting 2026 (AAS 247)*, Phoenix, Arizona, January 4-8.
2. **Zaman, MD. A. Al**, & Crossfield, Ian J. M. (2025). Towards precise Eta Earth Estimates: Reassessment of planet parameters and Statistical Validation for 144 Kepler Confirmed and Candidate Planets. *Great Lakes Exoplanet Area Meeting (GLEAM)*, University of Wisconsin-Madison, November 6-7.
3. **Zaman, MD. A. Al**, Crossfield, I. J. M., Brande, J., & Douglas, E. S. (2025). STERNA: A Python-based Batoid-integrated Observation Planning Tool for Exoplanet Science. *Adaptive Optics Summer School*, University of California Santa Cruz, August 10-15.

4. **Zaman, MD. A. Al**, Crossfield, I. J. M., & Brande, J. (2025). A New Eye in Space: Exploring the Potential of a Next-Generation Space Telescope for Exoplanet Science Cases. *KU Physics & Astronomy PALOOZA'25*, April 26.
5. **Zaman, MD. A. Al**, Islam, M. R., Maruf, H. M. A. R., & Nizam, Q. M. R. (2022). Study on the effectiveness of Kevlar and Polyethylene for radiation shielding in a spacecraft. *4th International Conference on Physics for Sustainable Development and Technology (ICPSDT-2022)*, January 22-23.
6. **Zaman, MD. A. Al**, Maruf, H. M. A. R., & Islam, M. R. (2021). Study on Shielding Effectiveness of a Conceptual Active Radiation Shield Constructed with Multi-Wall Carbon Nanotube Added YBCO-123. *National Conference on Physics*, Bangladesh Physical Society.
7. **Zaman, MD. A. Al**, et al. (2019). Space applications of High Tc Superconductivity and Superconducting Magnetic shield for manned space missions. *3rd ICPSDT-2019*, December 18-19.
8. **Zaman, MD. A. Al**, Ahmed, S., & Monira, N. J. (2018). An overview of Superconducting Magnetic Energy Storage (SMES) and Its Applications. *International Conference on Nanotechnology and Condensed Matter Physics (ICNCMP 2018)*.

SEMINAR TALKS

1. Toward Refined Eta Earth ($\eta_{\oplus}$) Constraints: Homogeneous Reanalysis of Planet Parameters and Statistical Validation of 188 Kepler HZ and Near-HZ Confirmed and Planet Candidates, *KU Astro-Space Seminar (Spring 2026)*, January 30, 2026
2. STERNA: A Python-Based Observation Planning Tool For Exoplanet Science, *KU Astro-Space Seminar (Fall 2025)*, August 29, 2025.

ASTRONOMICAL OBSERVING EXPERIENCE

W. M. Keck Observatory

Instrument: KPF / Mode: Remote

12 Half Nights

PROPOSALS (SUBMITTED)

1. PI - James Webb Space Telescope (JWST), Cycle 5

Title : Constraining Eta Earth: JWST Insights into a Super-Earth Orbiting a Solar Twin

AWARDS & HONORS

KU Physics and Astronomy Travel Support (\$750) <i>For: Attending 247th AAS Meeting in Phoenix, Arizona</i>	Nov 25, 2025
Best Faculty Member for Development and Growth <i>Occasion: National Textile Day, Northern University Bangladesh (NUB)</i>	Dec 12, 2023
Best Presenter (Oral Presentation) <i>5th ICPSDT-2023</i>	Sept 8, 2023
Best Poster Award (2 Awards) <i>4th ICPSDT-2022 and National Conference on Physics 2021</i>	2021, 2022
National Finalist: EATL-Prothom Alo Apps Challenge <i>For: Emergency Bangladesh Android App (Sponsor: World Bank)</i>	Feb 11, 2017

JOURNAL REVIEWER

- Nature Astronomy
- The European Physical Journal Plus (Springer)
- Journal of Radiation Research and Applied Sciences (Elsevier)

TEACHING (UNDERGRADUATE)

University of Kansas (KU)

<i>PHSX 216: General Physics I Laboratory (42 Students in 3 sections)</i>	Fall 2024
<i>ASTR 391: Introduction to Astronomy (Substitute for Prof. Crossfield)</i>	Fall'25 (1), Spring'26 (1)

Northern University Bangladesh (NUB)

<i>PHY 1305 & 1306: Electricity, Magnetism, Heat & Thermodynamics and Lab</i>	2019 – 2024
<i>EEE 3105 & 3106: Basic Electrical and Electronics Eng. and Lab</i>	2021
<i>PHY 1109 & 1110: Properties of Matter and Optics and Lab</i>	2019 – 2023

COMPUTER SKILLS

Programming:	Java, C, C++, Python
Software:	TOPCAT, DS9, MATLAB, SPENVIS (GEANT4), OLTARIS
Operating Systems:	Linux (Ubuntu), Windows
IDE:	Android Studio, Matlab, Apache NetBeans/Cordova

PROFESSIONAL RESPONSIBILITIES

REU Application Reviewer Panel Department of Physics and Astronomy, University of Kansas	Summer 2025&2026
Executive Committee Member Physics and Astronomy locally Organized Assembly (PALOOZA)	April 25, 2026
Judge (Poster) Undergrad Research Symposium, University of Kansas	April 25, 2025
Convener Seminar on Impact of Artificial Intelligence on Education and Industries, NUB	July 31, 2023
Co-Convener Celebration of National Textile Day, NUB	December 12, 2023
Convener Seminar on Contemporary Research in Dept. of Textile Engineering, NUB	November 12, 2022
Convener Workshop on MATLAB, Dept. of Textile Engineering, NUB	July 31, 2022
Convener: Promotion & Digital Activities Dept. of Textile Engineering, NUB	2023 – 2024
Recruitment Board Member (Lecturer) and Candidate Demo Organizer Office of Vice Chancellor and Human Resources, NUB	2023 – 2024
Co-Convener Celebration of National Textile Day, NUB	December 4, 2022
Technical Member Journal of Natural Science and Textile Technologies (JNSTT), NUB	2023 – 2024
Technical Member Textile Newsletter (Semesterwise), NUB	01/2020 – 06/2024
Advisor NUB Textile Research and Publications Club, NUB	2021 – 2023
Member Outcome Based Education Curriculum Formulation Central and Departmental Committee, NUB	2022
Member (Documentation) Central Admission Committee, NUB	2022 – 2024
Co-Convener Documentation Committee, Dept. of Textile Engineering, NUB	2021 – 2022
Executive Member Chittagong University Physics Alumni Reunion	2018
Executive Committee Member Celebration of Einstein's 100 Years of General Relativity Chittagong University Physics Society, Dept. of Physics, CU	2015

PROFESSIONAL TRAININGS

Summer School in Statistical Methodology for Astronomy <i>Center for Astrostatistics and Astroinformatics, Penn State University, USA</i>	June 1–5, 2026
Adaptive Optics Summer School <i>Center for Adaptive Optics (CFAO), UC Santa Cruz, USA</i>	Aug 11–15, 2025
Outcome Based Education (OBE) Curriculum Development <i>Northern University Bangladesh (NUB), Dhaka</i>	Jan 5–7, 2023

MENTORING

- 1. Joseph Discavage (2025 to present)** (Undergrad, University of Kansas)
Topic: Tools for exoplanet lightcurve fitting and limb darkening coefficients
- 2. Lasany Arefin Kunja (2022-2023)** (M.S., Chungnam National University, South Korea)
Paper: Effectiveness of radiation shields... (AIP Advances)
- 3. Sabyasachi Roy & Shorna Sarkar (2023)** (M. S., University of Dhaka, Bangladesh)
Paper: Comparative Analysis of Natural Polymers... (Materials Research Express)

OUTREACH

Guest Speaker, Astro Quest 2025 Organized by: Chittagong University Scientific Society, University of Chittagong	Oct 10, 2025
GRE and TOEFL Instructor Nextop-USA, (Higher Study Consultancy Center) (nextopusa.com)	01/2019 – 08/2019
Feature Writer <i>The Business Standard, Roar Media, Bigganjatra</i> Topics: Physics, Astronomy, Education, Technology, and Book Reviews. Profiles: Roar Media BigganJatra The Business Standard	30+ Articles
Personal Blog About: Research tools, literature review, and predatory practices. Link: Research Blog	Ongoing
Volunteer, Jamal Nazrul Islam Science Fair Demonstrated basic Physics experiments to elementary-level students.	2016
Examiner, Bangladesh Physics Olympiad Chittagong Chapter. Evaluated and graded competition answer scripts.	2016